Week 6 Progress Report

Interfacing with AI Models | CSR2: Evaluating LLM-Generated Transactions

Prepared by: Ben P | For: Dr. Ivanov | Date: October 14, 2026

Disclosure: This report was summarized by Claude (Anthropic) from my working session. The commands, runs, outputs and decisions described are mine; Claude helped organize and word them. Terminal output in Appendix A is copied verbatim from my PowerShell window.

1. Connection to the project goals

The team is evaluating whether an LLM can turn a student's request into a **correct registration transaction** (a DIDComm v2 message), and how to measure "correct". The harness scores each answer in four tiers and stops at the first failure: **syntactic** (valid JSON?), **schema** (right fields and type?), **consistency** (legal in the database: full sections, time conflicts?) and **semantic** (the right answer for what the student wanted?).

My role is **interfacing with the AI models and designing what we test**: choosing which LLMs to run (Task 1.3), categorizing the test cases by what they test (Task 1.4), and running cases through the team's test harness on different models. The harness itself is the team's shared work; I run it and extend the test design around it.

2. Summary of this week

- **Ran the team's draft test harness** (`run_semantic_eval.py`, OpenRouter backend) on all 6 input-format cases with Claude Sonnet 5: **6/6 semantic_ok**. The offline answer-key check also reports **6/6 cases sound**.
- **Swapped the model with one flag** on two cases (sem_001, con_002) across three models and got a different outcome in different tiers: a pass, a syntactic failure, a semantic failure and a consistency failure.
- **Task 1.3:** a justified list of 6 models with per-call cost and a ~$2.6 budget for 90 calls per model.
- **Task 1.4:** two criteria lists (17 syntactic, 41 semantic) and a map of all 36 existing test cases, with the gaps.
- **Checked every answer key by hand** in the input-format file and found two small data inconsistencies (Section 6).

3. Running implementation: what I ran and what happened

All runs use OpenRouter, temperature 0, one sample per case, JSON rendering unless noted. Full terminal output is in Appendix A.

3.1 Sonnet 5 on all six input-format cases

Case	Result	Detail
sem_001	semantic_ok	matches ground truth exactly
con_001	semantic_ok	matches ground truth exactly
con_002	semantic_ok	matches ground truth exactly
sem_002	semantic_ok	matches ground truth exactly
sem_003	semantic_ok	matches ground truth exactly
sem_004	semantic_ok	matches ground truth exactly

3.2 The same two cases on three models (model swap)

Model	sem_001	con_002	Cost for 2 calls
Claude Haiku 4.5	syntactic_error: invalid JSON (Expecting value: line 1 column 1)	syntactic_error: same	$0.0037
Claude Sonnet 5	semantic_ok	semantic_ok	$0.0116
Llama 3.1 8B Instruct	semantic_error: returned ['12345', '12346'], expected ['12345']	consistency_error: time conflict inside the cart, 30001 and 30002 on Monday	$0.0001

3.3 Earlier demo runs (Sonnet 5, same day)

Case	Rendering	Result	Cost
sem_001	json	semantic_ok (picked the Glassboro section, not Camden)	$0.0044
con_002	csv	consistency_error (time conflict: 30001 and 30002 on Monday)	$0.0043

3.4 Older cart-format cases on Haiku 4.5

cart-001, cart-002 and cart-005 (from `cart_test_cases.jsonl`) all returned syntactic_error with the same message (cost $0.0027 total). I read the raw output for sem_001 and con_002 only (Section 4); I have not yet opened the raw output of the cart run.

4. Findings

- **Haiku 4.5 wraps its answer in a markdown code fence.** The raw output for sem_001 is a JSON message inside ```` ```json ... ``` ````. The harness counts that as a syntactic error (README open question 2). On sem_001 the JSON inside the fence was correct (sections ["12345"]), so with fences stripped Haiku would have passed that case. On con_002 the JSON inside the fence was *wrong* (it returned all three sections 30001, 30002 and 30003), so it would still fail a later tier.
- **Llama 3.1 8B failed in two different ways.** On sem_001 it added a course the student never asked for (12346, MATH03000). On con_002 it chose two classes that overlap on Monday. The first is a semantic failure (the cart is legal but wrong); the second is caught by the consistency tier.
- **The tiers do what they are for.** The same two cases produced four different verdicts, which shows why a single pass/fail is not enough.
- **Input format may matter.** Sonnet 5 failed con_002 with the csv rendering (earlier run) and passed it with the json rendering. This is one sample per condition at temperature 0, so it suggests an effect but does not establish one.
- **Prompt note for the team:** the prompt contract does not say "output only JSON, no markdown". Whether to keep fenced output as an error or strip fences before parsing is an open team decision; I have not changed the shared prompt.

5. Task 1.3: LLMs to test

Chosen to support four comparisons: closed vs open weights; size (Haiku / Sonnet / Opus and Llama 8B / 70B); cross-vendor (Claude, GPT, Llama); and local vs cloud (Llama 8B on Ollama and on OpenRouter).

#	Model	Type	Why included	Cost per call	90 calls
1	anthropic/claude-opus-5.5	Closed	Upper bound. $4 / $20 per M tokens	~$0.02 (measured)	~$1.80
2	anthropic/claude-sonnet-5	Closed	Mid tier, demo model. ~$3 / $15 per M (implied from measured cost)	$0.0059 (measured)	$0.53

#	Model	Type	Why included	Cost per call	90 calls
3	anthropic/claude-haiku-4.5	Closed	Cheap tier. $1 / $5 per M. Wraps JSON in code fences (Section 4)	$0.0017 (measured)	$0.15
4	openai/gpt-4o-mini	Closed	Different vendor. $0.128 / $0.597 per M (weighted average paid)	~$0.00026 (estimate)	~$0.02
5	meta-llama/llama-3.1-8b-instruct	Open	Open weights; local vs cloud check. ~$0.02 / $0.05 per M (third-party)	$0.000025 (measured)	~$0.002
6	Llama 3.1 70B (exact slug to be confirmed)	Open	Larger open model, size effect vs 8B. $0.85 / $0.85 per M (weighted average paid)	~$0.0012 (estimate)	~$0.11

Estimates assume about 1,200 input and 175 output tokens per call. Total for 90 calls on each of the six models is about $2.6; Opus is about 70% of it. Rows 4 and 6 use weighted-average prices actually paid, not list prices. The Sonnet 5 price is implied from measured cost, and the Llama 70B slug still needs to be checked on openrouter.ai/models. Optional: Claude Sonnet 5.5 (thinking is always on, so its cost should be measured first).

6. Task 1.4: Test case types

Syntactic criteria can be checked from the model's output alone. **Semantic** criteria need the student's request and record to know whether the answer is right.

6.1 Syntactic tests (17)

Group	Criteria
Format	S1 valid JSON; S2 no code fence or markdown wrapper (seen: Haiku 4.5 on every answer I read, and Sonnet 5 on cart-006 per the README); S3 no prose before or after; S4 exactly one object; S5 not empty, truncated or a refusal
Message structure	S6 has id, type, body; S7 exact type URI; S8 body.student_id present; S9 body.sections is a list even for one course (targeted by cart-003); S10 sections are strings; S11 no extra fields, correct types
Values (checked against the schedule)	S12 hallucinated section ID; S13 altered student_id; S14 invented course value; S15 duplicate section; S16 full section chosen; S17 time conflict inside the cart

Coverage: every case exercises S1-S11 by default and cart-003 targets S9. There are no cases that target the other format checks directly. Proposed fix: hand-written malformed outputs fed straight to the checker, with no model call needed.

6.2 Semantic tests (41)

Group	Criteria (existing cases that cover them; GAP = none yet)
Temporal reasoning	time-of-day words (reg-sem-001/002, reg-001, reg-004); earlier/later (reg-sem-005/006, reg-005, cart-010); hard cutoff such as "no 8 a.m." or "ends by 5 PM" (cart-006, sem_004); partial overlap (con_002, cart-005); chained conflicts (cart-009). GAP: boundary values, back-to-back sections, sections with several meetings, AM/PM and noon traps
Days	MW vs TR (reg-sem-003/004, cart-002); shared day with a different pattern (con_002). GAP: day exclusion ("no Fridays")
Capacity	avoid a full section (con_001, cart-004); compare open seats (cart-008); right answer not listed first (con_001, cart-001, cart-004). GAP: last seat (29/30)
Course and section identity	look-alike IDs (cart-007); pick A not B (reg-sem-007/008, reg-003); negation (reg-sem-013). GAP: course vs section vs CRN mix-up, same title with a different ID
Request constraints	modality (sem_002, reg-sem-009/010); instructor (reg-sem-011/012); campus from residence, implicit (sem_001); several constraints (reg-sem-014, cart-006, sem_004); hard vs soft (reg-sem-015); quantity (sem_003, sem_004, reg-002). GAP: impossible request with no valid answer, ambiguous request

Group	Criteria (existing cases that cover them; GAP = none yet)
Student record	do not retake a passed course (sem_003, sem_004); repeatable course, prerequisite met, in-progress course not enough, transfer credit counts, requirement options, requirement already satisfied, department filter (all sem_004). GAP: W vs failed, minimum grade, any_of/all_of rules, credit-load limit
Completeness	includes every requested course (cart set); adds nothing unrequested (cart-001); several valid answers accepted (sem_003, sem_004)

Coverage was mapped from the case descriptions, the README, and the cases I read in full (sem_001, sem_003, sem_004); the "GAP" items are the safest claims. Cases reg-001 to 005 and reg-sem-013 to 015 were classified from their request text, not their answer keys.

6.3 All 36 existing cases by primary rule

Rule	Count	Rule	Count
A1 capacity	4	A7 instructor	2
A2 time conflict	3	A8 credit load	0
A3 prerequisite (alone)	0	A9 compound	3
A4 program requirement	1	A10 section preference (time/days)	12
A5 campus	1	A11 baseline / format / course selection / look-alike IDs	6
A6 modality	3	A12 negation	1

By tier: about 30 target semantic, 5 consistency (con_001, con_002, cart-004, cart-005, cart-009) and 1 schema (cart-003). The 36 cases are 6 in `input_format_cases.json`, 5 in `test_cases.jsonl`, 15 in `semantic_test_cases.jsonl` and 10 in `cart_test_cases.jsonl`.

6.4 Gaps and proposals

- Hidden (implicit) rules appear in only 3 of 36 cases (sem_001, sem_003, sem_004). These matter most for the research question.
- Preference cases are a third of the set (12 of 36); prerequisites alone and credit load have no cases, and program requirement has one.
- No case where no valid answer exists. Only the 6 input-format cases use a full student record. Only sem_003 and sem_004 have more than one correct answer.
- The cart cases were written by Claude and have not been checked by a teammate (per the README).
- Proposal: add hand-written bad outputs for the syntactic list and about a dozen semantic cases for the GAP items.

7. Data notes on the test-case file

I recomputed the correct answer for all six cases in `input_format_cases.json` by hand (seats, overlaps, prerequisites, the 5 PM cutoff, repeatable courses) and they match the answer keys. Two small inconsistencies for the case owner:

- **sem_003** has no prerequisites list, so CS02435 is a correct answer there but wrong in sem_004 (its prerequisite CS04114 is still in progress).
- **sem_004**: CS04113 is 4 credits on the transcript but 3 credits in the schedule. This does not change the answer.

8. Teamwork and coordination

I missed the team meeting this week. I told the team about my work on Tasks 1.3 and 1.4, and I am sharing these results and the Haiku code-fence finding in the group chat. I am also asking for a short joint run-through of the harness before class, and for a team decision on fenced output (README open question 2). The harness, the prompt contract and the test cases are the team's shared work; I ran them unchanged, and my contribution is the model list, the test-case categories, and the runs and findings above.

9. Limitations

- One sample per case at temperature 0, so these results show the harness works and the tiers separate failures; they are not pass rates and do not rank the models.
- The demo model comparison covers two cases only (sem_001 and con_002).
- In Table 5, GPT-4o-mini and Llama 70B costs are estimates, the Sonnet 5 price is implied, and the Llama 70B slug is unconfirmed.
- I have not yet opened the raw output of the Haiku cart run, so the cause of those three failures is presumed (same fence) and not confirmed.

10. Next steps

- Run all 36 cases on the model list in Table 5 and report rates with confidence intervals, plus cost and latency per model.
- Compare json, csv and nl renderings on more samples to test whether input format affects accuracy.
- Agree with the team on fenced output, then re-score Haiku.
- Add malformed-output fixtures for the syntactic list and new cases for the semantic gaps, starting with hidden-rule cases.

11. Proof attached to this submission

Item	What it shows
Screenshot: `--check` output (6/6 cases sound)	answer keys verified offline
Screenshot: Sonnet 5 on all 6 cases (6/6 semantic_ok)	harness running end to end
Screenshot: three-model swap on sem_001 + con_002	four different tier outcomes
Screenshot: Haiku cart run (0/3 syntactic_error)	older case format also runs
Result files: all6_sonnet.json, swap_*.json (3), cart_haiku.json, demo.json, demo2.json	raw outputs, verdicts, tokens and costs
Screenshot of the Teams cards for 1.3 and 1.4, and of my Discord/Teams message	task ownership and team communication
GitHub: link to my commit on my branch	files uploaded to the team repository

Appendix A: terminal output (copied verbatim from PowerShell)

Commands are shown wrapped for readability. In my window `python` was not on the path, so I used the Python launcher `py`.

```
$ py run_semantic_eval.py --test-file input_format_cases.json --check
sem_001: ok
con_001: ok
con_002: ok
sem_002: ok
sem_003: ok
sem_004: ok

6/6 cases sound.

$ py run_semantic_eval.py --backend openrouter --model anthropic/claude-sonnet-5
      --test-file input_format_cases.json --rendering json --output all6_sonnet.json
sem_001: semantic_ok          | matches ground truth exactly
con_001: semantic_ok          | matches ground truth exactly
con_002: semantic_ok          | matches ground truth exactly
sem_002: semantic_ok          | matches ground truth exactly
sem_003: semantic_ok          | matches ground truth exactly
sem_004: semantic_ok          | matches ground truth exactly

$ foreach ($m in "anthropic/claude-haiku-4.5","anthropic/claude-sonnet-5","meta-llama/llama-3.1-8b-instruct")
  { py run_semantic_eval.py --backend openrouter --model $m --test-file input_format_cases.json
      --task-ids sem_001 con_002 --rendering json --output ("swap_" + ($m -replace '/','_') + ".json") }
sem_001: syntactic_error      | invalid JSON (Expecting value: line 1 column 1 (char 0))
con_002: syntactic_error      | invalid JSON (Expecting value: line 1 column 1 (char 0))

0/2 semantic_ok on anthropic/claude-haiku-4.5. Cost $0.0037. Full results in swap_anthropic_claude-haiku-4.5.json
sem_001: semantic_ok          | matches ground truth exactly
con_002: semantic_ok          | matches ground truth exactly

2/2 semantic_ok on anthropic/claude-sonnet-5. Cost $0.0116. Full results in swap_anthropic_claude-sonnet-5.json
sem_001: semantic_error       | field mismatch vs. ground truth: {'sections': {'expected': ['12345'], 'actual': ['12345', '12346']}}
con_002: consistency_error    | time conflict inside the cart: 30001 and 30002 on Monday

0/2 semantic_ok on meta-llama/llama-3.1-8b-instruct. Cost $0.0001. Full results in swap_meta-llama_llama-3.1-8b-instruct.json

$ py run_semantic_eval.py --backend openrouter --model anthropic/claude-haiku-4.5
      --test-file cart_test_cases.jsonl --task-ids cart-001 cart-002 cart-005 --output cart_haiku.json
cart-001: syntactic_error      | invalid JSON (Expecting value: line 1 column 1 (char 0))
cart-002: syntactic_error      | invalid JSON (Expecting value: line 1 column 1 (char 0))
cart-005: syntactic_error      | invalid JSON (Expecting value: line 1 column 1 (char 0))

0/3 semantic_ok on anthropic/claude-haiku-4.5. Cost $0.0027. Full results in cart_haiku.json

--- Earlier demo runs, Sonnet 5 ---
$ python run_semantic_eval.py --backend openrouter --model anthropic/claude-sonnet-5
      --test-file input_format_cases.json --task-ids sem_001 --rendering json --output demo.json
sem_001: semantic_ok          | matches ground truth exactly
1/1 semantic_ok on anthropic/claude-sonnet-5. Cost $0.0044. Full results in demo.json

$ python run_semantic_eval.py --backend openrouter --model anthropic/claude-sonnet-5
      --test-file input_format_cases.json --task-ids con_002 --rendering csv --output demo2.json
con_002: consistency_error    | time conflict inside the cart: 30001 and 30002 on Monday
0/1 semantic_ok on anthropic/claude-sonnet-5. Cost $0.0043. Full results in demo2.json
```

Raw output of the Haiku sem_001 answer (from swap_anthropic_claude-haiku-4.5.json, field raw_output):

```
```json
{"type": "https://example.org/course-registration/1.0/register", "id": "a7f2c9e1-3b5d-4f8a-9e2c-1d4b6a8f3c5e",
 "body": {"student_id": "S001", "sections": ["12345"]}}
```
```

Raw output of the Haiku con_002 answer:

```
```json
{"type": "https://example.org/course-registration/1.0/register", "id": "a7f2c9e1-5b3d-4e8a-9c22-1d4e5f6a7b8c",
 "body": {"student_id": "S001", "sections": ["30001", "30002", "30003"]}}
```
```